POWER / DEFAULT-OFF LED SWITCH

MCU / DUAL DATA / SERVICE

PASSIVE CV THRU / SENSING / ADC

U1
STM32G031G8U6TR

U2
AP63205 fixed 5V

U3
TLV75533

U4
TPS2552DBVR-1

U5
SN74LVC2T45

J3
DUAL LED G / S / L / R

U6
OPA197 unity

U7
TLV9001 unity

D3
BAS70-04

Q1
MMBT3904

D1
PMEG4010 +rail

D2
PMEG4010 -rail

R14
330 1%

R15
330 1%

R16
100k 1%

J1
CV IN

J2
CV THRU

SW1
ON / PROGRAM / CV

L1
4.7uH SPM4020-LR

J5
POWER + / G / -

R11
68k 1%

R17
100k 1%

R18
100k 1%

R19
100k 1%

R1
10M 1%

R2
10M 1%

R3
82k 1%

R12
100k 1%

R13
10k 1%

R20
4.7k 1%

R21
10k 1%

C7
4.7u

C8
100n

R4
82k 1%

R5
82k 1%

R6
82k 1%

R22
10k 1%

R23
100k 1%

C1
10u

C9
100n

C10
100n

C14
100n

R7
1k 1%

R8
1k 1%

R9
10k 1%

C2
22u

C3
22u

C4
100n

TP1
+3V3

TP2
GND

TP3
SWDIO

R10
10k 1%

R24
330 1%

R25
1k 1%

C5
4.7u

C6
4.7u

C15
10u

TP4
SWCLK

TP5
NRST

C11
100n

C12
100n

C13
10n

C16
10u

#FLG01
Power source declaration

#FLG02
Power source declaration

C17
100n

#FLG03
Power source declaration

#FLG04
Power source declaration

#FLG05
Power source declaration

#FLG06
Power source declaration

Identical net labels are connected. U6 senses through 20 Mohm: U7 buffers the ADC.
ADC=(100/135)*(VDD/2+CV/8). TPS2552-1 uses active-low EN: PA5 HIGH enables via NPN.
PJ-3582 pins 4/5 detect insertion and are unused. Pin 6 TIP directly connects IN to THRU.
Power transients, MLCC bias, fault heat and connector continuity require hardware tests.

Engineering draft. Controller is not routed. No hardware qualification claimed.

Light Block

Sheet: /
File: controller.kicad_sch

Title: Light Block A5 controller

Size: A2	Date: 2026-11-06	**Rev: A5**
KiCad E.D.A. 10.0.8		Id: 1/1